COS 101- Introduction To Computing Science

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LEARNING OUTCOMES

- ✓ **explain** basic components of computers and other computing devices;
- ✓ **describe** the various applications of computers;
- ✓ **explain** information processing and its roles in the society;
- ✓ **describe** the Internet, its various applications and its impact;
- ✓ **explain** the different areas of the computing discipline and its specializations; and
- ✓ **demonstrate** practical skills on using computers and the internet.

SECTION 1: UNDERSTANDING THE COMPUTER

1.1 BASIC CONCEPTS

1.1.1 Definition of a Computer

A computer is an electronic device that accepts raw data as input, processes it based on instructions (programs) and produces useful output (information).It is not just a machine but a system that integrates **hardware**, **software**, **people** and **processes** to solve problems. It can perform a wide range of tasks: calculations, data storage, communication, control and automation.

A COMPUTER SYSTEM

1.1 BASIC CONCEPTS

1.1.2 Data and Information

Data: Raw, unprocessed facts such as numbers, text, symbols, or sounds (e.g., 23, blue, #1500). Data by itself may not have meaning.

Information: Processed, organized, or structured data that conveys meaning and is useful for decision-making (e.g., “The student scored 23 marks in Physics”).

1.1 BASIC CONCEPTS

1.1.3 Characteristics of a Computer

- ✓ Speed: Computers can perform millions of operations per second (measured in MHz, GHz).
- ✓ Accuracy: Errors only occur due to wrong input or faulty programming; otherwise, computers are precise.
- ✓ Automation: Once programmed, computers can perform tasks without human intervention.
- ✓ Storage: Computers can store vast amounts of data in compact devices (from kilobytes to petabytes).

1.1 BASIC CONCEPTS

- ✓Versatility: They can perform different tasks, from word processing to complex simulations.
- ✓Connectivity: Computers can connect globally through networks (Internet), enabling communication and information sharing.
- ✓Diligence: Unlike humans, computers do not suffer from fatigue or boredom; they can work continuously.
- ✓Programmability: Computers can be reprogrammed to perform different tasks.

1.1 BASIC CONCEPTS

1.1.4 Input–Process–Output (IPO) Model

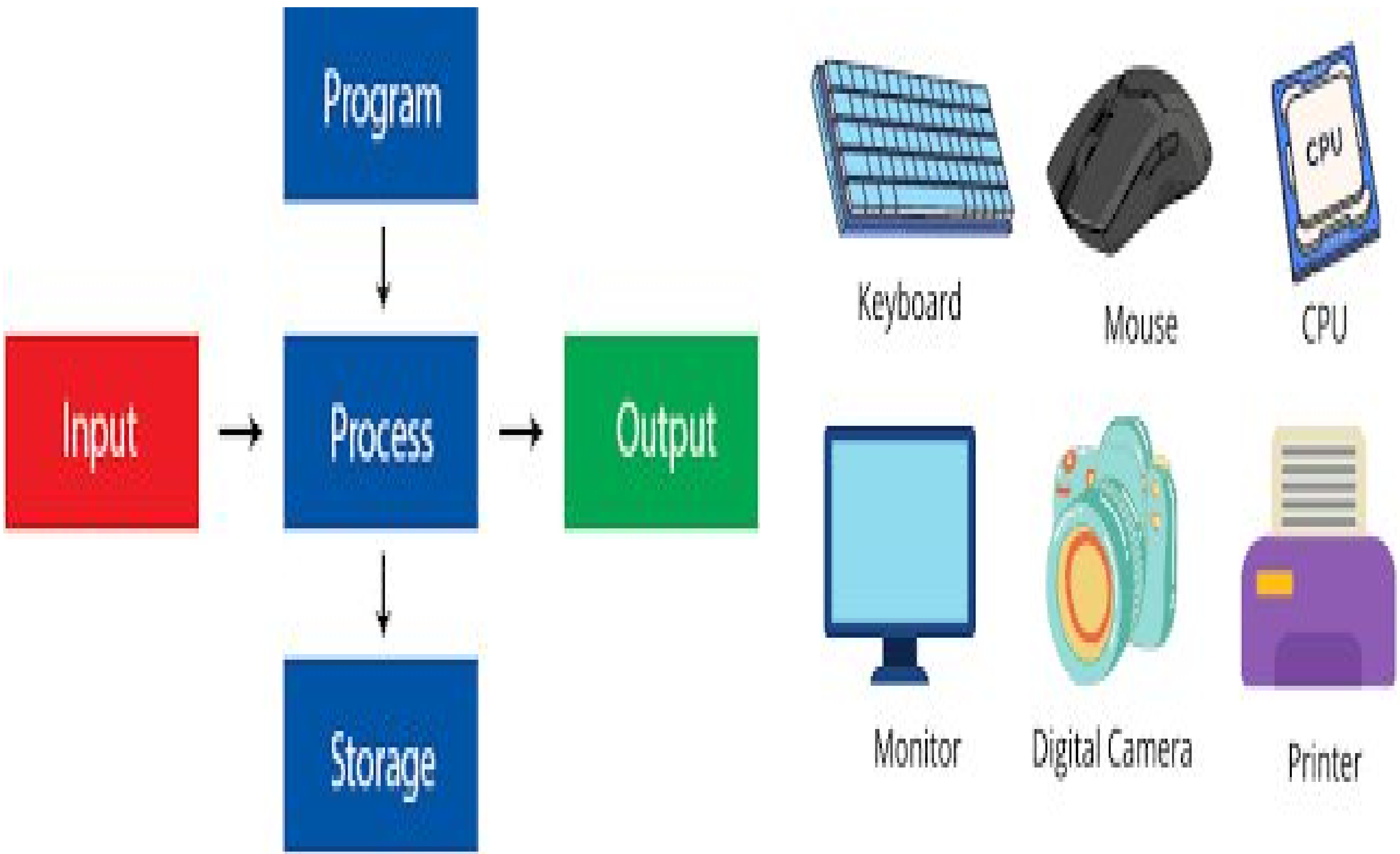
This model describes how computers function:

Input: Data and instructions entered via devices like keyboards, scanners, or sensors.

Process: CPU executes instructions to transform input data into meaningful results.

Output: Processed information is displayed via monitors, printers or speakers.

IPO MODEL With Examples



1.1 BASIC CONCEPTS

The function of a CPU (Central Processing Unit) is to execute instructions from programs, perform calculations, and manage the flow of data within a computer, acting as its "brain". It retrieves instructions from memory, performs arithmetic and logic operations, and controls other hardware components to process data and produce output.

In summary, Executes Instructions, Performs calculations, Controls Data flow, Processes Data, Manages System Operations

1.2 HISTORICAL REVIEW OF THE COMPUTER

1.2.1 Early Tools and Pre-Computing Devices

Before the electronic computer, humans used mechanical and manual devices for counting and calculations. Examples include:

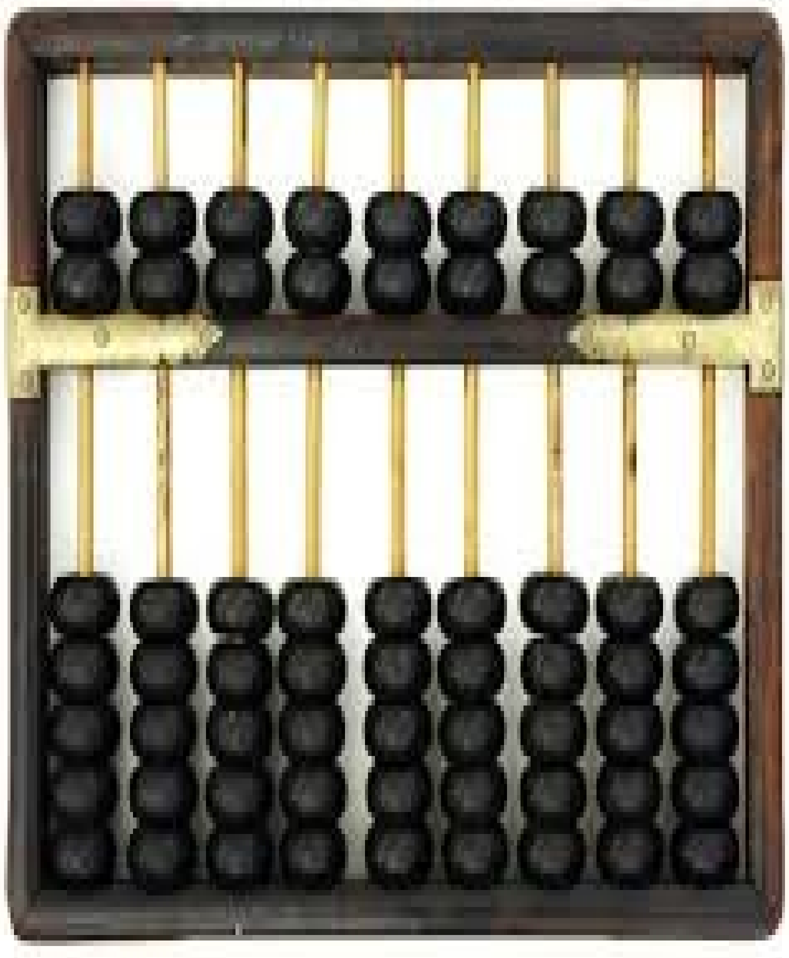
Abacus (c. 2400 BC): An ancient counting frame invented in Mesopotamia, widely used in Asia and still taught in some places for arithmetic.

Napier's Bones (1617): Invented by John Napier; used rods inscribed with multiplication tables to simplify arithmetic operations.

Pascal's Adding Machine (1642): Blaise Pascal created a mechanical calculator that could add and subtract.

Leibniz's Step Reckoner (1673): Improved calculator by Gottfried Wilhelm Leibniz, capable of multiplication and division.

ABACUS – Calculating Device/Napier's Bones



Pascal Adding Machine/Leibniz’s Reckoner

1.2 HISTORICAL REVIEW OF THE COMPUTER

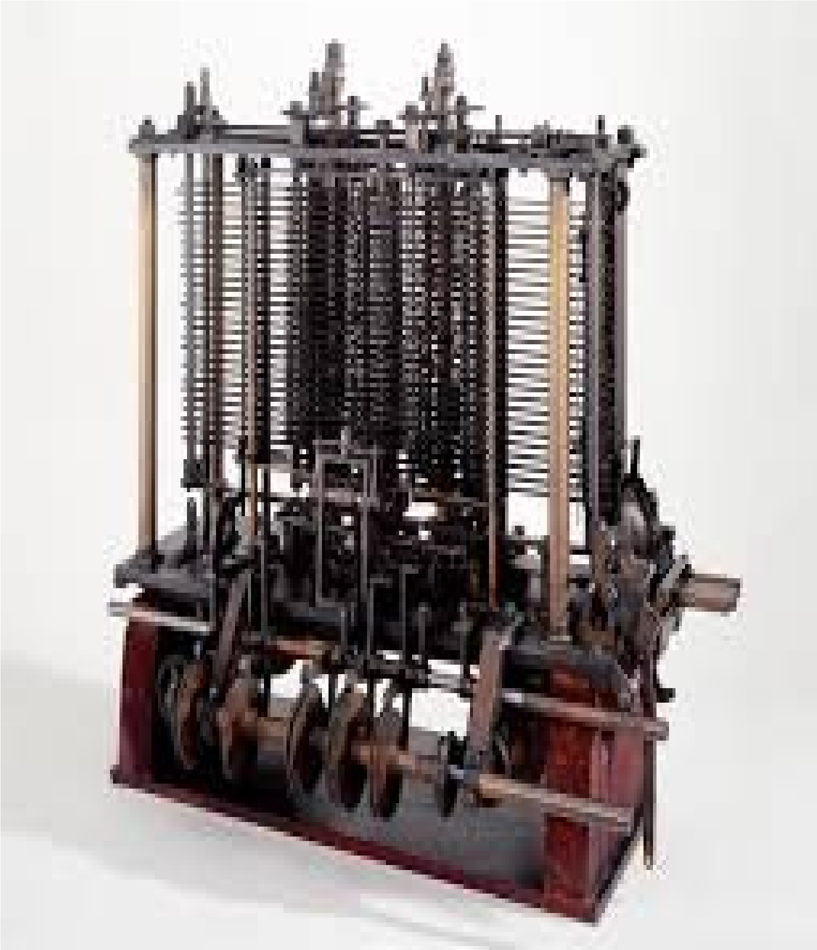
Jacquard Loom (1801): Joseph Jacquard's weaving machine controlled by punched cards, an early concept of programmability.

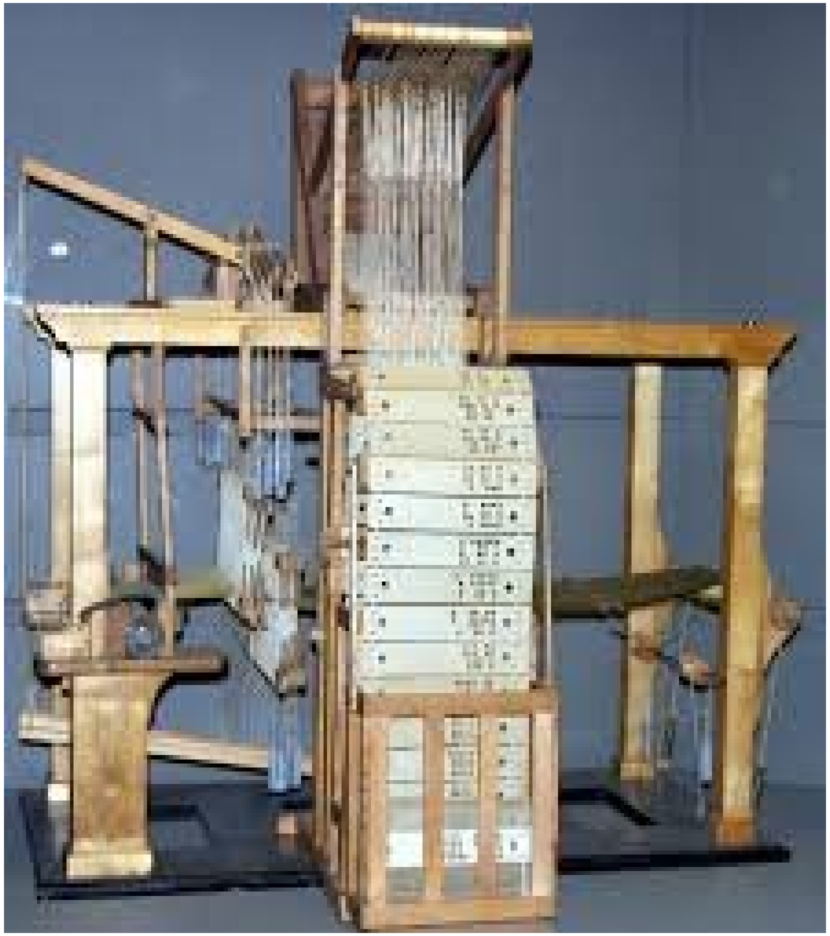
Charles Babbage's Analytical Engine (1837): Designed as the first "general-purpose" programmable machine; though never completed, it inspired later computers.

Ada Lovelace: Recognized as the world's first computer programmer for her work on Babbage's design.

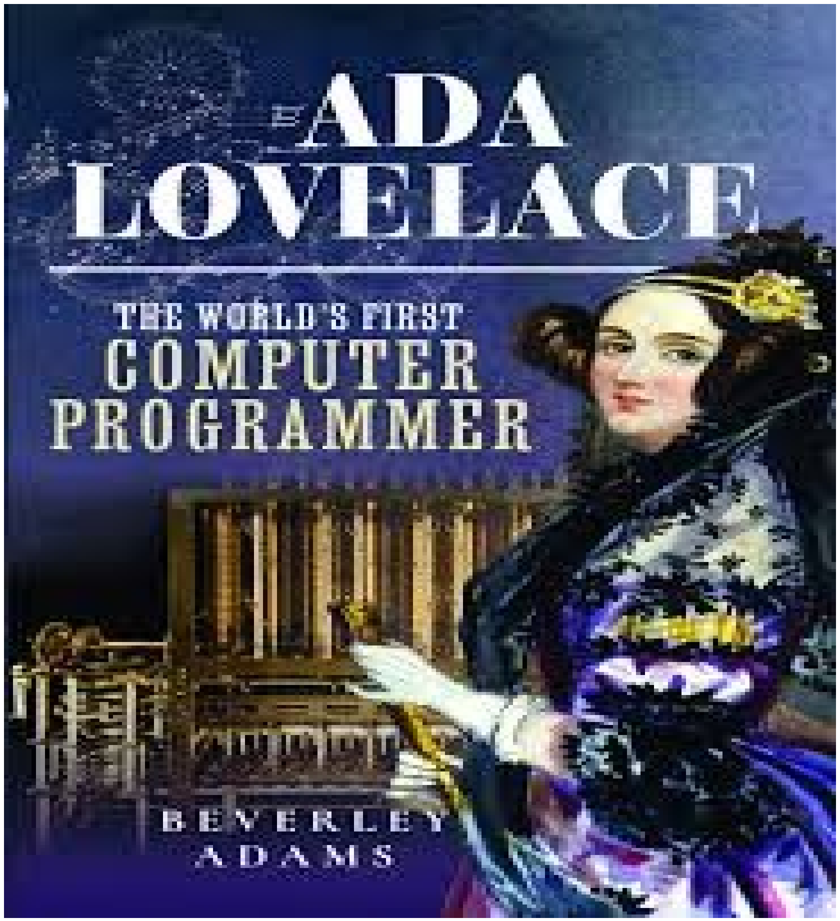
Herman Hollerith's Tabulating Machine (1890): Used punched cards to process U.S. census data; led to the formation of IBM.

Charles Babbage’s Analytical Engine/ Jacquard Loom





Ada Lovelace/ Herman Hollerith’s Tabulating Machine



1.2 HISTORICAL REVIEW OF THE COMPUTER

1.2.2 Electromechanical Era (1930s–1940s)

These were transitional devices that combined mechanical parts with electrical power:

Konrad Zuse’s Z3 (1941): Considered the first programmable digital computer.

Harvard Mark I (1944): An electromechanical calculator developed by IBM and Harvard University.

Konrad Zuse's Z3/Harvard Mark I

1.2.3 Generations of Electronic Computers

First Generation (1940–1956): Vacuum Tube Computers

Technology: Vacuum tubes for circuitry and magnetic drums for memory. Examples: ENIAC, EDVAC, UNIVAC I, IBM 701.

- **Characteristics:**

✓ Large in size, consumed huge electricity.

✓ Very expensive and generated lots of heat.

✓ Input/Output: Punch cards and paper tape.

✓ Programming in machine language and assembly.

ENIAC/EDVAC

IBM 701 / UNIVAC 1

1.2.3 Generations of Electronic Computers

Second Generation (1956–1963): Transistor Computers Technology: Transistors replaced vacuum tubes. Examples: IBM 1401, IBM 7090.

❑ **Characteristics:**

- ✓ More reliable, smaller, faster, and cheaper.
- ✓ Used magnetic core memory.
- ✓ Programming with early high-level languages like FORTRAN and COBOL.
- ✓ Used in business, science, and engineering.

IBM 1401/IBM 7090

1.2.3 Generations of Electronic Computers

Third Generation (1964–1971): Integrated Circuit Computers Technology: Integrated Circuits (ICs) replaced transistors. Examples: IBM 360 Series, PDP-8.

- **Characteristics:**

- ✓ Increased speed, efficiency and reliability.
- ✓ Minicomputers introduced.
- ✓ Multiple applications could run simultaneously (multiprogramming).
- ✓ Input/Output: Keyboards, monitors and operating systems.

IBM 360/PDP-8

1.2.3 Generations of Electronic Computers

Fourth Generation (1971–1980s): Microprocessor Computers Technology: Microprocessors (entire CPU on a single chip). Examples: Intel 4004, IBM PC, Apple II.

❑ **Characteristics:**

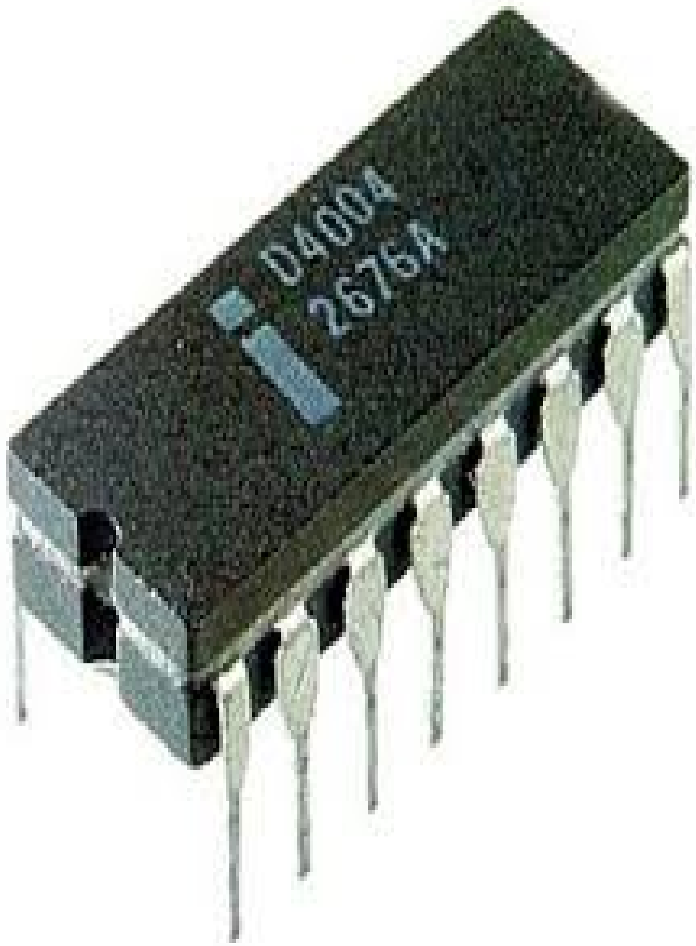
✓ Personal computers became affordable and popular.

✓ Graphical user interfaces (GUIs) and computer networks emerged.

✓ Large-scale integration (LSI) and Very Large-scale integration (VLSI).

✓ Software industry grew rapidly.

INTEL 4003/APPLE II



1.2.3 Generations of Electronic Computers

Fifth Generation (1980s–Present): AI and Beyond

Technology: VLSI, parallel processing, AI, and advanced networking. Examples: Modern PCs, smartphones, cloud computing platforms.

- **Characteristics:**

- ✓ Focus on Artificial Intelligence (AI) and machine learning.
- ✓ Natural language processing, robotics, and expert systems.
- ✓ Internet and mobile computing dominate.
- ✓ Cloud computing and edge computing offer on-demand resources.
- ✓ Quantum computing research is advancing.

Smartphones/Cloud Computing Platform

1.3 CLASSIFICATION OF COMPUTERS

Computers may be classified according to different criteria such as purpose, data handling, size/capacity, and generation. This classification helps students appreciate the diversity of computing devices and their applications.

1.3.1 CLASSIFICATION BY PURPOSE

- **General-Purpose Computers**

✓ Designed to perform a variety of tasks.

✓ Can be reprogrammed to suit different applications (e.g., word processing, spreadsheets, gaming).

Examples: Personal computers (laptops, desktops), servers.

Server/Desktop Computer

1.3 CLASSIFICATION OF COMPUTERS

- **Special-Purpose Computers**
- ✓ Built to perform one specific task or a narrow range of tasks.
- ✓ Optimized for efficiency in that function.

Examples: ATMs, traffic light control systems, washing machine processors, embedded systems in cars.

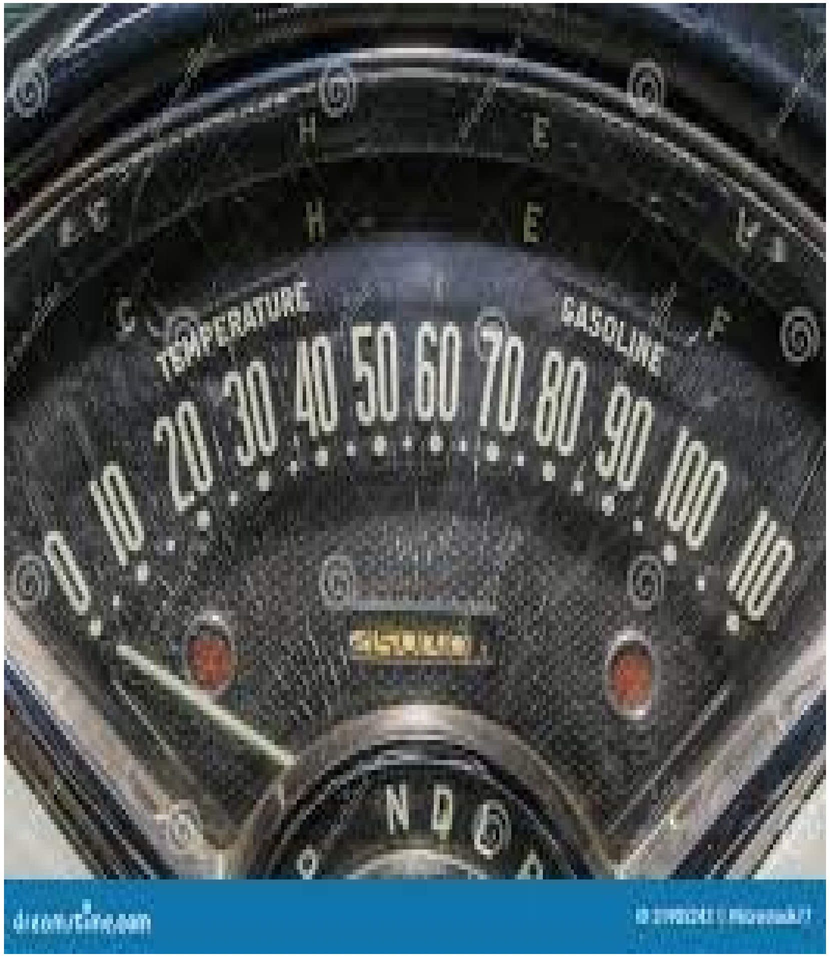
ATMS/Traffic Light Control Systems/Washing Machine Processor

1.3.2 CLASSIFICATION BY DATA HANDLING

- **Analog Computers**
- ✓ Process data in continuous form (signals, waves).
- ✓ Used in scientific applications like measuring temperature, pressure or voltage.

Examples: Old speedometers, early flight simulators.

Early Flight Simulator/Old Speedometer



1.3.2 CLASSIFICATION BY DATA HANDLING

- **Digital Computers**
- ✓ Process data in discrete form (binary 0s and 1s).
- ✓ Most common type today—found in PCs, smartphones, servers.
- ✓ Perform logical, arithmetic, and decision-making tasks.

Decision Making Process/ALU Operations

C_1	C_0	Arithmetic $C_2 = 0$	Logical $C_2 = 1$
0	0	Addition	OR
0	1	Subtraction	AND
1	0	Multiplication	NOT
1	1	Division	EX-OR

1.3.2 CLASSIFICATION BY DATA HANDLING

❑ **Hybrid Computers**

✓ Combine features of analog and digital systems.

✓ Used in areas where both continuous data measurement and discrete calculations are needed.

Examples: Hospital monitoring systems (measure heartbeat as analog, convert to digital for analysis).

Patient Monitoring Systems

1.3.3 CLASSIFICATION BY SIZE AND CAPACITY

Computers differ in physical size, processing power and storage capacity.

- **Supercomputers**

- ✓ Most powerful, fastest, and most expensive computers.

Perform trillions of calculations per second

- ✓ Used for weather forecasting, climate modeling, nuclear simulations, AI research.

Examples: Fugaku (Japan), Summit (USA).

Fugaku /Summit Supercomputers

1.3.3 CLASSIFICATION BY SIZE AND CAPACITY

❑ Mainframe Computers

- ✓ Large systems used by big organizations for massive data processing.
- ✓ Support thousands of users simultaneously.
- ✓ Used in banks, insurance companies, government agencies.

Examples: IBM Z-series.

Mainframe Computers

1.3.3 CLASSIFICATION BY SIZE AND CAPACITY

- **Minicomputers (Midrange Computers)**
  - ✓ Smaller than mainframes but larger than microcomputers.
  - ✓ Serve small to medium businesses for departmental processing.

Examples: PDP-11, VAX series (historical), modern midrange servers.

IBM Midrange Server/Minicomputer

1.3.3 CLASSIFICATION BY SIZE AND CAPACITY

❑ **Microcomputers (Personal Computers)**

✓ Most common and affordable computers today.

✓ Designed for individual users.

Subcategories:

Desktop Computers: Stationary, powerful systems for offices and homes.

Laptop Computers: Portable PCs with rechargeable batteries.

Tablets: Touchscreen-based portable systems.

Smartphones: Mobile computers with telecommunication features.

Desktop Computer/ Laptop Computers

Tablets/Smartphones

1.3.3 CLASSIFICATION BY SIZE AND CAPACITY

Workstations

- ✓ High-performance computers designed for technical/scientific applications.
- ✓ Used by engineers, architects, graphic designers.
- ✓ Have more processing power than PCs but less than mainframes.

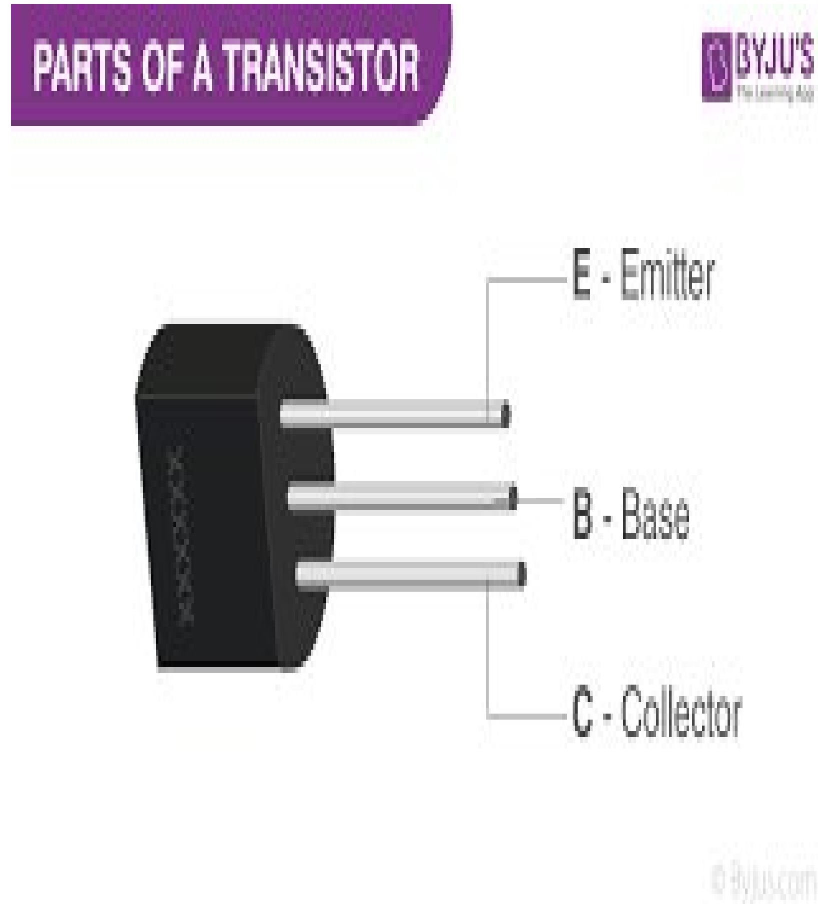
Workstations

1.3.4 CLASSIFICATION BY GENERATION

Computers can also be classified based on the technology used in their construction:

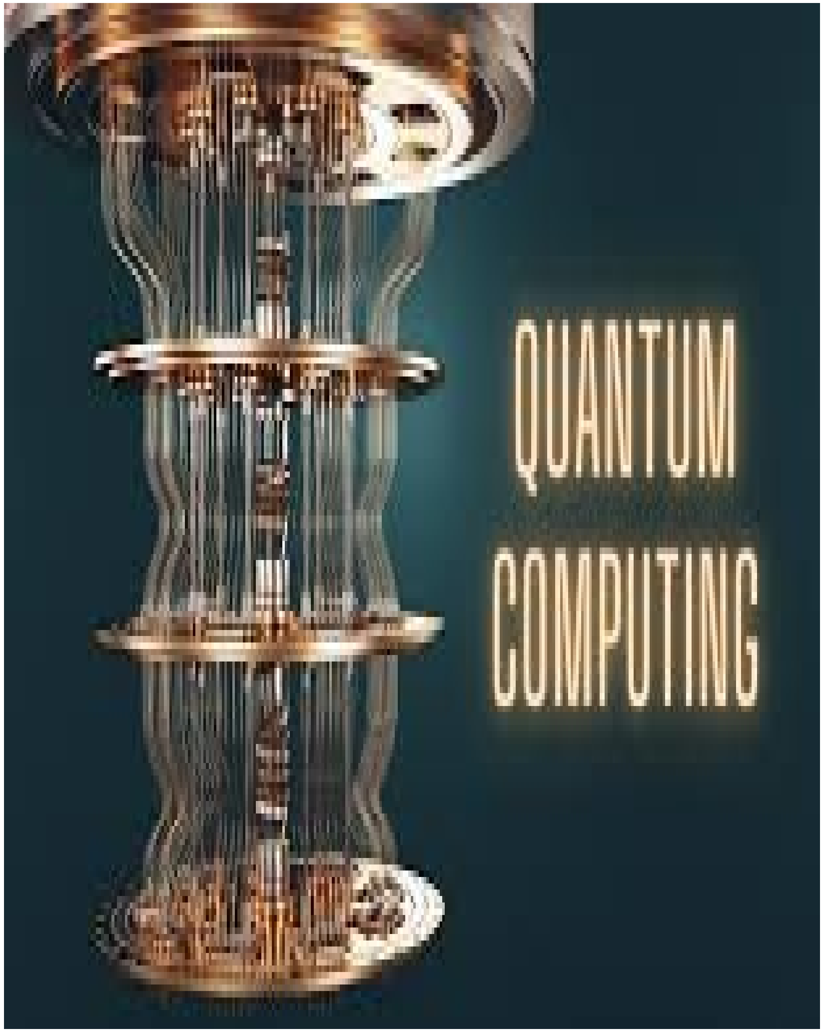
- ✓ **First Generation**: Vacuum tubes (1940s–1950s).
- ✓ **Second Generation**: Transistors (1950s–1960s).
- ✓ **Third Generation**: Integrated Circuits (1960s–1970s).
- ✓ **Fourth Generation**: Microprocessors (1970s–1980s).
- ✓ **Fifth Generation**: AI, parallel processing, quantum computing (1980s–present).

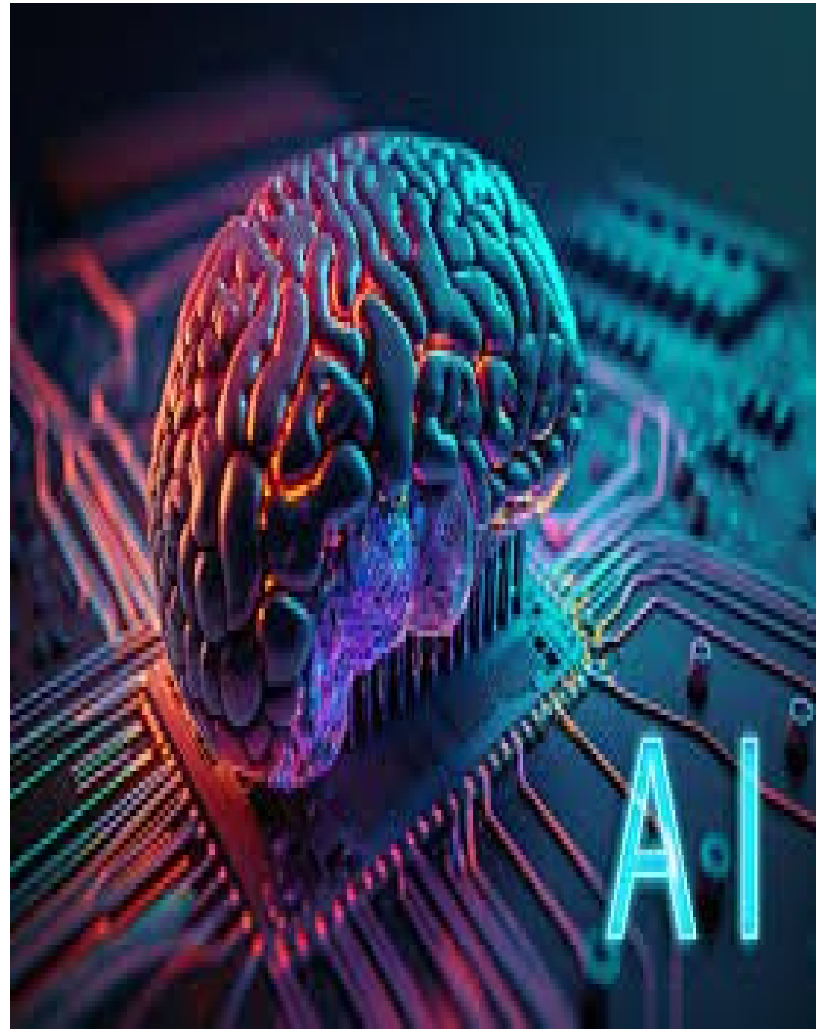
Vacuum Tubes/Transistors



Integrated Circuits/Microprocessors

Quantum Computing/AI





1.3.5 OTHER MODERN CLASSIFICATIONS

- ✓ **Wearable Computers**: Smart watches, fitness trackers.
- ✓ **Embedded Computers**: Microcontrollers in appliances, cars, medical devices.
- ✓ **Network Computers / Thin Clients**: Depend on central servers for most processing.
- ✓ **Cloud-Based Systems**: Computing resources delivered over the Internet

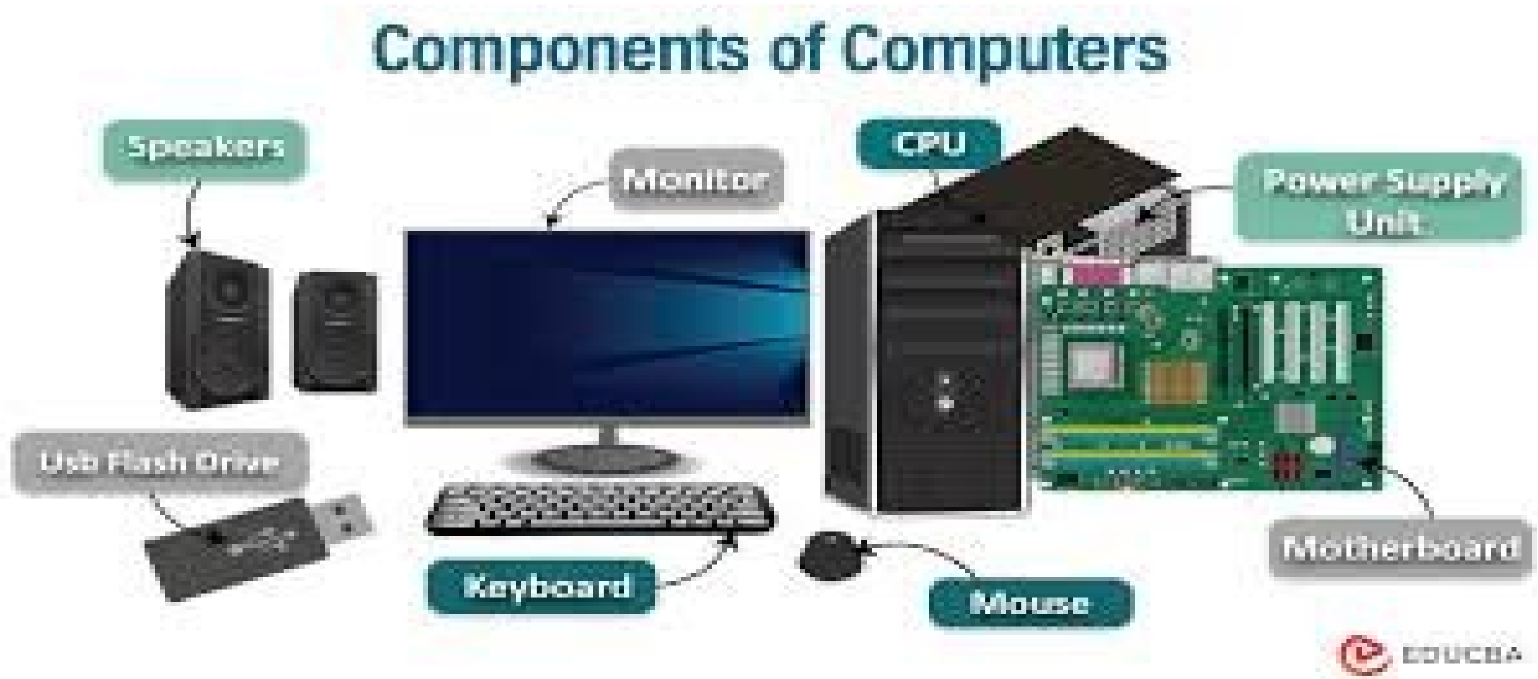
Fitness Tracker/Smart Watches

Network Computer/Cloud-Based Systems

1.4 COMPUTER HARDWARE

- ✓ Computer hardware refers to the physical components of a computer system that are essential for performing input, processing, storage, and output tasks.
- ✓ Hardware interacts with software to execute instructions and produce results.

Computer Hardware Components



1.4.1 MAJOR COMPONENTS OF COMPUTER HARDWARE

- **Central Processing Unit (CPU)** – The “brain” of the computer:

- ✓ **Control Unit (CU):** Directs data flow between input, output, memory and CPU.
- ✓ **Arithmetic Logic Unit (ALU):** Performs arithmetic operations (+, -, /, *) and logical operations (comparisons, decisions).
- ✓ **Registers:** Small, high-speed storage locations for temporary data during processing.
- ✓ **Cache Memory:** High-speed memory close to the CPU to speed up processing.
- ✓ **Clock:** Synchronizes operations within the CPU.

1.4.1 MAJOR COMPONENTS OF COMPUTER HARDWARE

- **Input Devices** – Hardware used to provide data and instructions to the computer:
- ✓ **Keyboard**: Standard and ergonomic types for text input.
- ✓ **Mouse/Touchpad**: Pointing devices for navigation.
- ✓ **Scanner**: Converts physical documents/images into digital form.
- ✓ **Microphone**: Inputs audio data.
- ✓ **Digital Cameras/Webcams**: Capture visual data.
- ✓ **Sensors/IoT Devices**: Measure environmental variables or physical conditions.

1.4.1 MAJOR COMPONENTS OF COMPUTER HARDWARE

- **Output Devices** – Hardware that displays or presents processed information:

- ✓ **Monitors/Displays**: CRT, LCD, LED, OLED.
- ✓ **Printers**: Inkjet, laser, dot matrix for paper output.
- ✓ **Speakers/Headphones**: Audio output.
- ✓ **Projectors**: Large-scale display for presentations.
- ✓ **Haptic Devices**: Provide tactile feedback (e.g., vibrations in gaming).

LED 20’’Monitors/HP LaserJet Printer

Audio-Output Speaker/Projector/Headphone

1.4.1 MAJOR COMPONENTS OF COMPUTER HARDWARE

- **Storage Devices** – Components that store data temporarily or permanently:

✓ **Primary Storage (Volatile):**

Random Access Memory (RAM): Stores data and instructions temporarily while programs run.

Read-Only Memory (ROM): Stores permanent instructions like BIOS/firmware.

✓ **Secondary Storage (Non-Volatile):**

Hard Disk Drives (HDD), *Solid-State Drives (SSD)*

Optical Discs (CD/DVD/Blu-ray)

✓ **Tertiary Storage**: Backup and archival solutions, e.g., tape drives, cloud storage.

1.4.1 MAJOR COMPONENTS OF COMPUTER HARDWARE

- **Motherboard (System Board)** – The main circuit board connecting CPU, memory, storage and peripherals:

- ✓ **Slots and Ports:** For expansion cards, USB devices, network connections.
- ✓ **Chipset:** Manages data flow between CPU, memory, and peripherals.
- ✓ **Power Connectors:** Supply electricity to components.

1.4.1 MAJOR COMPONENTS OF COMPUTER HARDWARE

- **Peripheral Devices** – Additional devices that expand functionality:
  - ✓ **External Drives**: USB hard drives, external SSDs.
  - ✓ **Networking Devices**: Routers, switches, network interface cards (NICs).
  - ✓ **Specialized Devices**: Graphic tablets, 3D printers, VR/AR headsets.

1.5: COMPUTER SOFTWARE

Computer software refers to the set of instructions, data, or programs that direct a computer to perform specific tasks.

- ✓ Unlike hardware, which is the tangible component, software is intangible and serves as the ***"intelligence"*** of a computer system.
- ✓ Without software, the hardware is non-functional.

A. DEFINITION AND CHARACTERISTICS OF SOFTWARE

- **Definition:** Software is a collection of programs and associated documentation that enables a computer to perform useful work.

- **Key Characteristics:**

- ✓ Intangible – cannot be physically touched.
- ✓ Developed, not manufactured – unlike hardware, software is designed and written by programmers.
- ✓ Prone to errors and bugs – requires testing and debugging.
- ✓ Evolves over time – can be updated or modified.
- ✓ Runs on hardware platforms – must be compatible with operating systems and computer architecture.

B. CATEGORIES OF SOFTWARE

Computer software is broadly divided into System Software and Application Software.

✓ **System Software**

This is software designed to manage and control the computer hardware so that application software can function. It acts as a bridge between user applications and hardware.

❑ **Types of System Software:**

Operating System (OS):

Examples: Windows, Linux, macOS, Android.

B. CATEGORIES OF SOFTWARE

- ✓ **Functions:** Process management, memory management, device management, file system handling, user interface and security.
- ✓ **Utility Programs:**
- ✓ Special-purpose programs that enhance system performance.

Examples: Antivirus, Disk Cleanup, Backup utilities, File compression tools. ***Popular antivirus examples include Norton, McAfee, Bitdefender, Kaspersky, Avast, TotalAV, and Malwarebytes***

- ✓ **Device Drivers:**

Specialized programs that allow the OS to communicate with hardware devices like printers, scanners, and network cards.

CATEGORIES OF SOFTWARE

✓ Application Software

This refers to programs designed for end-users to perform specific tasks.

❑ Types of Application Software:

General-Purpose Application Software: Examples: Microsoft Office (Word, Excel, PowerPoint), Browsers (Chrome, Firefox), Media Players.

Special-Purpose Application Software:

Designed for specific industries or users. Examples: AutoCAD (engineering), SPSS (statistics), MATLAB (scientific computing), QuickBooks (accounting).

Web and Mobile Applications: Examples: WhatsApp, Zoom, Gmail, Facebook.

C. DIFFERENCES BETWEEN SYSTEM AND APPLICATION SOFTWARE

Aspect	System Software	Application Software
Purpose	Manages hardware and provides platform	Helps users perform specific tasks
Examples	Windows OS, Linux, Device Drivers	MS Word, Photoshop, Zoom
User Interaction	Runs in background, less direct interaction	Direct interaction with users
Dependency	Required for computer functionality	Depends on system software to run

D. PROGRAMMING SOFTWARE (Third Category)

In modern computing, a third category is often recognized:

- ✓ **Programming Software:** Tools used by developers to create other software.

Examples: Compilers, Interpreters, Debuggers, Integrated Development Environments (IDEs) like Eclipse, PyCharm, Visual Studio.

E. SOFTWARE DEVELOPMENT PROCESS

- ✓ **Requirement Analysis** – understanding user needs.
- ✓ **Design** – outlining system structure.
- ✓ **Coding/Implementation** – writing source code.
- ✓ **Testing and Debugging** – identifying and fixing errors.
- ✓ **Deployment** – installing and making the software operational.
- ✓ **Maintenance and Updates** – continuous improvements, bug fixing, and security patches.

1.6: PROGRAMMING THE COMPUTER

A. DEFINITION

Computer programming is the process of designing, writing, testing, debugging, and maintaining instructions (programs) that tell a computer how to carry out tasks.

B. KEY CONCEPTS IN PROGRAMMING

1. **Program** – A sequence of instructions written to perform a specific task.

2. **Programming Language** – A formal language used to write computer programs (e.g., Python, Java, C++).

3. **Source Code** – The human-readable version of a program.

1.6: PROGRAMMING THE COMPUTER

4. **Object Code** – The machine-readable (binary) translation of the source code.

5. **Compiler/Interpreter** – Tools that convert source code into machine code for execution.

6. **Algorithm** – A step-by-step procedure for solving a problem.

C. LEVELS OF PROGRAMMING LANGUAGES

Programming languages are classified into levels of abstraction:

1. **Machine Language** (1st Generation)

Written in binary (0s and 1s).Directly executed by the CPU. Extremely fast but difficult for humans to read/write.

C. LEVELS OF PROGRAMMING LANGUAGES

2. **Assembly Language** (2nd Generation)

Uses symbolic codes (mnemonics) like ADD, SUB, MOV. Easier than machine language but requires an assembler to convert to machine code.

3. **High-Level Languages** (3rd Generation)

Closer to human language, easy to learn. Examples: Fortran, COBOL, C, C++, Java, Python.

Require a compiler or interpreter.

4. **Very High-Level Languages** (4th Generation)More abstract, often used for databases, report generation. Examples: SQL, MATLAB, R.

C. LEVELS OF PROGRAMMING LANGUAGES

5. **Modern Languages** (5th Generation and beyond).Used in artificial intelligence, machine learning, and advanced problem-solving. Examples: Prolog (logic-based), Python (AI & ML applications).

6. **Event-Driven Programming** – Execution is triggered by events (e.g., JavaScript, visual Basic).

SECTION 2:THE INTERNET,ITS APPLICATIONS AND ITS IMPACTS ON THE WORLD TODAY2.1: INTRODUCTION TO THE INTERNET

A. INTRODUCTION

- ✓ The Internet is a global network of interconnected computers that communicate with each other using standardized protocols.
- ✓ It is often referred to as the “network of networks”, linking millions of private, public, academic, business and government networks worldwide.

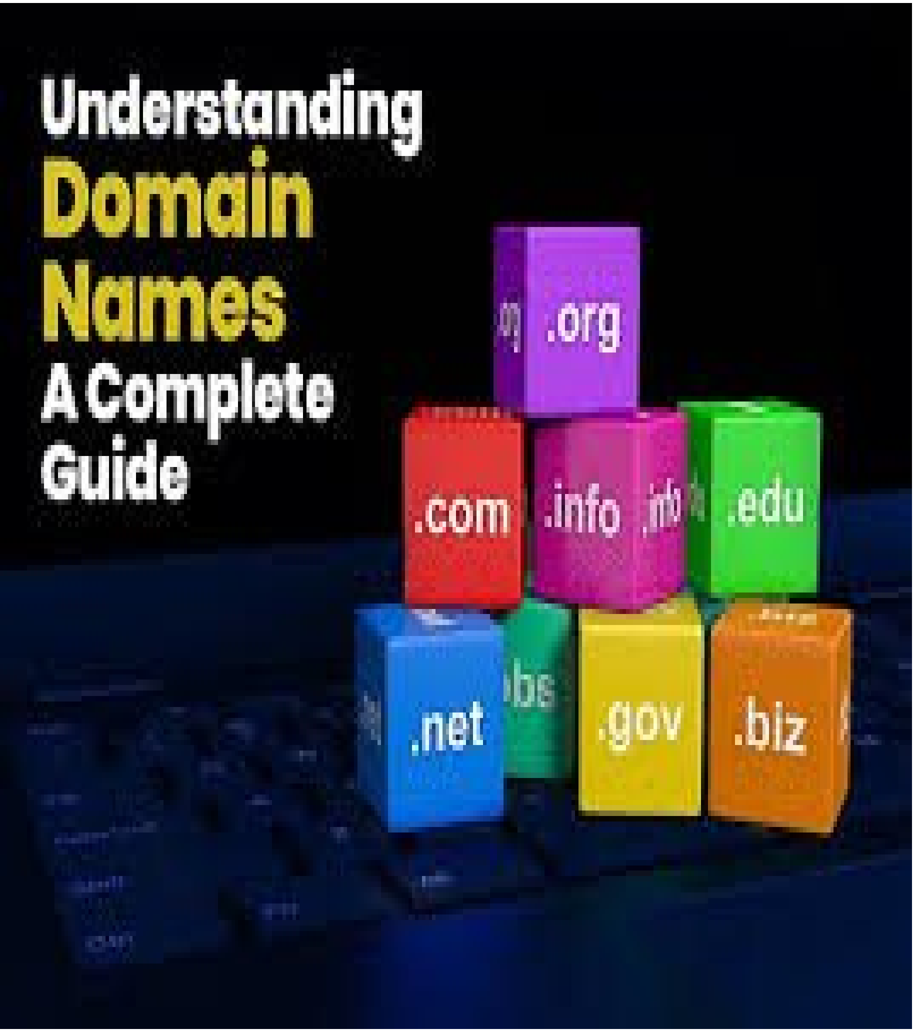
B. DEFINITION OF KEY TERMS

- ✓ **Internet** – A global system of interconnected computer networks that use the Transmission Control Protocol/Internet Protocol (TCP/IP) to communicate.
- ✓ **World Wide Web (WWW)** – A service on the Internet that allows access to information via websites and hyperlinks.
- ✓ **ISP (Internet Service Provider)** – A company that provides Internet access (e.g., MTN, Glo, AT&T).
- ✓ **IP Address** – A unique number that identifies a device on the Internet.
- ✓ **Domain Name** – A human-readable Internet address (e.g., www.veniteuniversity.edu.ng).

Internet and World Wide Web

Examples of Internet Service Providers in Nigeria

Domain Name/IP Address



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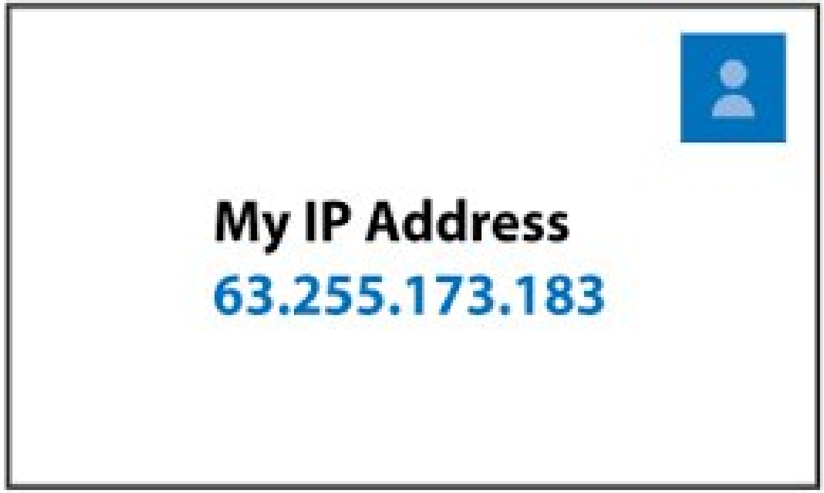


C. HISTORY OF THE INTERNET (Brief Review)

- ✓ **1960s:** Concept of networking computers started with ARPANET (Advanced Research Projects Agency Network) in the USA.
- ✓ **1970s:** Development of TCP/IP protocols.
- ✓ **1980s:** Universities and research institutions connected; email became popular.
- ✓ **1990s:** World Wide Web invented by Tim Berners-Lee; Internet became commercialized.
- ✓ **2000s** – Present: Explosive growth in mobile Internet, cloud computing, e-commerce, and social media.

D. CHARACTERISTICS OF THE INTERNET

- ✓ **Global Connectivity** – Access information and communicate worldwide.
- ✓ **Decentralized System** – No single owner or control authority.
- ✓ **Interoperability** – Devices of different types can connect and exchange data.
- ✓ **Accessibility** – Available to anyone with a device and Internet connection.
- ✓ **Multimedia Support** – Supports text, audio, video, and interactive content.

E. BASIC REQUIREMENTS TO CONNECT TO THE INTERNET

- ✓ **Computer or Smart Device** – Laptop, smartphone, or tablet.
- ✓ **Modem/Router** – Device that connects to an ISP.
- ✓ **ISP Subscription** – Access provided by service providers.
- ✓ **Browser Software** – Chrome, Firefox, Safari, Edge.
- ✓ **Network Protocols** – TCP/IP for communication.

A. INTRODUCTION

- ✓ The World Wide Web (WWW) is one of the most widely used services on the Internet.
- ✓ To access and interact with web content, users require a web browser—a software application that retrieves, interprets, and displays web pages.
- ✓ Web navigation refers to the process of moving from one webpage or website to another using hyperlinks, menus, or search engines.
- ✓ Together, browsers and navigation tools make the Internet practical, interactive, and user-friendly.

B. DEFINITION OF KEY TERMS

1. **Web Browser** – A software application used to access and view information on the World Wide Web (e.g., Chrome, Firefox, Safari).

2. **Webpage** – A single document on the Internet, usually written in HTML and accessible via a browser.

3. **Website** – A collection of related web pages under one domain (e.g., www.veniteuniversity.edu.ng).

4. **URL** (Uniform Resource Locator) – The unique address of a resource on the Internet.

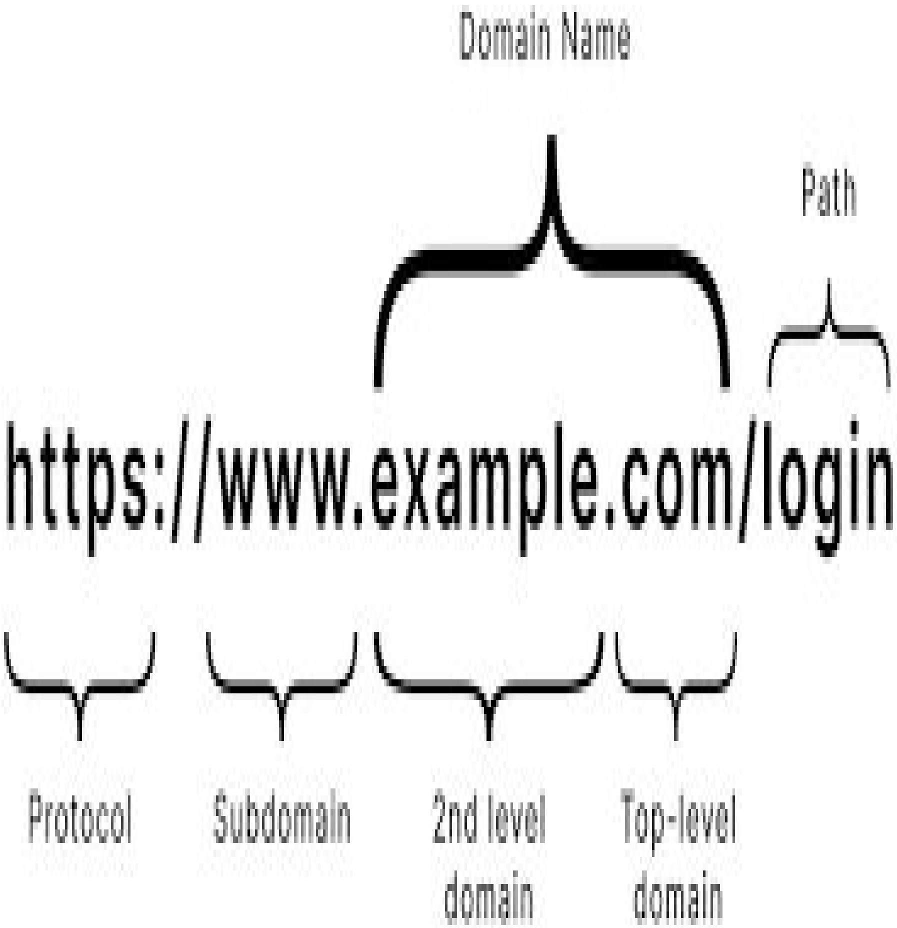
5. **Hyperlink** – A clickable text, image, or icon that connects to another webpage.

6. **Search Engine** – A platform that indexes and retrieves information (e.g., Google, Bing, Yahoo).

Examples of Web Browser Firefox/Chrome/Safari

Webpages/Website

Domain Name



Domain Name
Path
https://www.example.com/login
Protocol
Subdomain
2nd level domain
Top-level domain

HOSTINGER
Subdomain
www.yourdomain.com
Domain
Subdomain
news.yourdomain.com
Domain

Hyperlink/Search Engine

C. EXAMPLES OF POPULAR INTERNET BROWSERS

- ✓ **Google Chrome** – Known for speed, security, and wide usage.
- ✓ **Mozilla Firefox** – Open-source, customizable, strong privacy features.
- ✓ **Safari** – Default browser for Apple devices.
- ✓ **Microsoft Edge** – Successor to Internet Explorer, integrated with Windows.
- ✓ **Opera** – Lightweight browser with built-in VPN and ad blocker

2.3: E-MAIL AND ONLINE COMMUNICATION

B. DEFINITION OF KEY TERMS

1. **E-mail (Electronic Mail)** – A method of exchanging digital messages via the Internet.

2. **E-mail Address** – A unique identifier for sending/receiving e-mails (e.g., student@veniteuniversity.edu.ng).

3. **Inbox** – The folder where incoming messages are stored.

4. **CC (Carbon Copy)** – Sends a copy of the email to additional recipients.

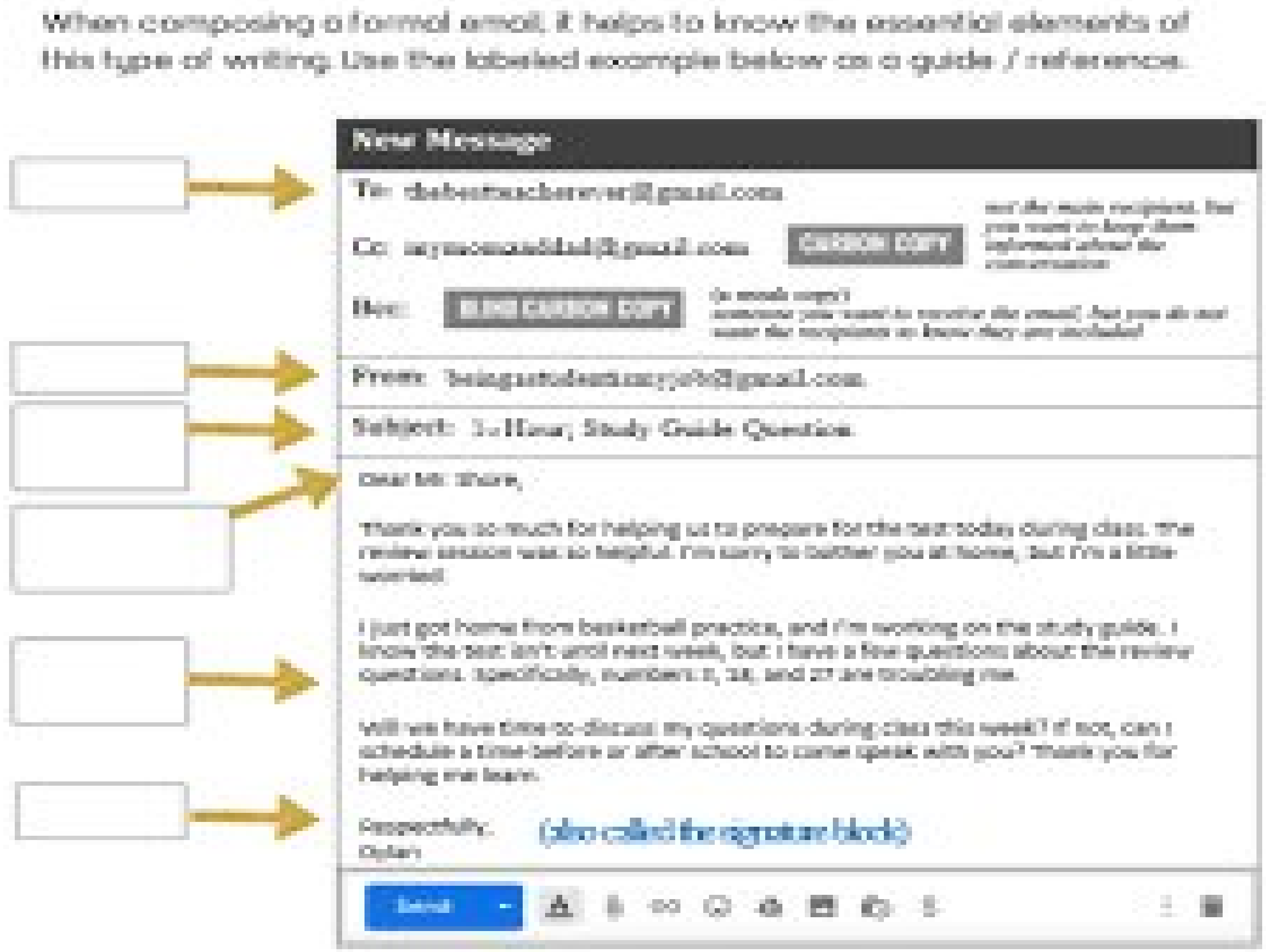
5. **BCC (Blind Carbon Copy)** – Sends a copy of the email without other recipients knowing.

6. **Attachment** – A file (e.g., document, image) sent along with an e-mail.

7. **Spam** – Unsolicited, often irrelevant or harmful messages.

8. **Online Communication** – Exchange of information through Internet-based platforms (e.g., WhatsApp, Zoom).

Components of an Email



C. COMPONENTS OF AN E-MAIL MESSAGE

1. **Header** – Includes sender’s address, recipient’s address, subject and timestamp.

2. **Subject Line** – A short summary of the email content.

3. **Greeting/Salutation** – e.g., Dear Prof. Kolawole.

4. **Body** – The main content of the message.

5. **Closing** – e.g., Sincerely, Yours Faithfully.

6. **Signature** – Contact details or professional identity of the sender.

7. **Attachments** – Documents, images, or files included with the email.

E-mail header
Greeting
Text
Signature
Body
From
Alex Diaz
Date
To
Subject
bean dip
Cc
Hi Guys,
--Alex
Alex T. Diaz
332 Toast Lane
East Providence,

D. STEPS IN USING AN E-MAIL

1. **Creating an Account** – Register with an email service provider (e.g., Gmail, Yahoo, Outlook).

2. **Composing a Message** – Write the subject and body, add recipients.

3. **Attaching Files** – Add documents, images, or other files if necessary.

4. **Sending the Email** – Transmit the message instantly.

5. **Receiving and Reading** – The recipient opens the email from their inbox.

6. **Replying or Forwarding** – Responding directly or sharing the email with others.

7. **Organizing Messages** – Use folders, labels, or search to manage e-mails.

E. ADVANTAGES OF E-MAIL

1. **Speed** – Messages are delivered almost instantly.
2. **Cost-effectiveness** – Cheaper than traditional postal services.
3. **Accessibility** – Can be accessed anytime, anywhere.
4. **Documentation** – Provides a written record of communication.
5. **File Sharing** – Supports attachments of documents, images, and multimedia.
6. **Global Reach** – Connects people worldwide without barriers.

F. LIMITATIONS OF E-MAIL

1. **Spam and Junk Mail** – Clutters inbox and may contain harmful links.

2. **Phishing Attacks** – Fraudulent emails used to steal information.

3. **Overload** – Large volume of emails can be overwhelming.

4. **Requires Internet Access** – Cannot be used offline.

5. **Impersonal Nature** – Lacks the emotional richness of face-to-face communication.

2.4: CLOUD COMPUTING AND INTERNET APPLICATIONS

- ✓ **Cloud computing** is the delivery of computing services—including storage, servers, databases, networking, software, analytics and artificial intelligence—over the **Internet** ("the cloud") rather than through local hardware or personal computers.
- ✓ It allows individuals, organizations and institutions to access and use powerful computing resources on-demand **without physically owning** or **maintaining them.**

Cloud computing is typically characterized by:

- ✓ On-demand self-service (resources can be provisioned automatically).
- ✓ Broad network access (available from anywhere via the Internet).
- ✓ Resource pooling (shared infrastructure serving multiple users).
- ✓ Rapid elasticity (resources can scale up or down as needed).
- ✓ Measured service (pay-as-you-go model).

Types of Cloud Computing

1. **Infrastructure as a Service (IaaS)**

Provides virtualized computing resources like servers, networks and storage.

Users rent infrastructure but manage their own applications and operating systems.

Example: Amazon Web Services (AWS) EC2, Microsoft Azure VM, Google Compute Engine.

2. **Platform as a Service (PaaS)**

Provides a platform for developers to build, test, and deploy applications without worrying about managing infrastructure. Example: Google App Engine, Microsoft Azure App Services, Heroku.

Types of Cloud Computing

3. **Software as a Service (SaaS)**

Provides complete software applications delivered via the Internet.Users access software through browsers without installation.

Example: Gmail, Microsoft 365, Zoom, Dropbox.

4. **Other Emerging Models**

Function as a Service (FaaS) or Serverless Computing – run code without managing servers (e.g., AWS Lambda).Backend as a Service (BaaS) – backend APIs, databases, and storage for mobile/web apps

Deployment Models of Cloud

1. **Public Cloud** – services offered over the public Internet and available to anyone (e.g., AWS, Azure, Google Cloud).

2. **Private Cloud** – dedicated cloud infrastructure used by a single organization.

3. **Hybrid Cloud** – combination of public and private clouds to balance flexibility, security, and cost.

4. **Community Cloud** – shared cloud infrastructure for specific groups with common goals (e.g., universities, research centers).

Advantages of Cloud Computing

- ✓ **Cost Efficiency** – reduces need for expensive hardware and software purchases.
- ✓ **Scalability** – resources can grow or shrink based on demand.
- ✓ **Accessibility** – users can work from anywhere with Internet access.
- ✓ **Collaboration** – teams can work on shared platforms in real-time.
- ✓ **Reliability & Backup** – cloud providers offer disaster recovery and data backup.
- ✓ **Automatic Updates** – software and infrastructure are kept up-to-date by the provider.

Disadvantages of Cloud Computing

- ✓ **Internet Dependence** – requires strong and reliable connectivity.
- ✓ **Security Concerns** – risks of data breaches, cyberattacks, and unauthorized access.
- ✓ **Privacy Issues** – data stored on third-party servers may raise legal/regulatory concerns.
- ✓ **Downtime Risks** – provider outages may affect availability of services.
- ✓ **Hidden Costs** – pay-as-you-go can become expensive without proper monitoring.

2.5: EXPLORING ONLINE SERVICES

Definition of Online Services

- ✓ ***Online services*** are platforms, applications, or systems delivered over the Internet that provide users with specific functions, resources, or benefits.
- ✓ They are designed to ***make*** communication, information access, commerce, entertainment, education and productivity easier, faster, and more efficient.
- ✓ Simply put, online services ***connect*** people, businesses, and governments ***through*** the Internet, allowing users to perform tasks ***without*** being physically present.

CATEGORIES OF ONLINE SERVICES

1. **Communication Services:**

- ✓ **Email** (Gmail, Yahoo Mail, Outlook).
- ✓ **Instant messaging** (WhatsApp, Telegram, Messenger).
- ✓ **Video conferencing** (Zoom, Google Meet, Microsoft Teams).
- ✓ **Social networking** (Facebook, Instagram, Twitter/X, LinkedIn).

2. **Information and Search Services**

- ✓ **Search engines** (Google, Bing, DuckDuckGo).
- ✓ **Online encyclopedias and knowledge platforms** (Wikipedia, Quora).
- ✓ **News and media websites** (BBC, CNN, Punch, Vanguard).

CATEGORIES OF ONLINE SERVICES

3. **Financial Services (E-Banking & Fintech)**

- ✓ **Internet banking platforms** (GTBank Internet Banking, Zenith e-banking).
- ✓ **Mobile money services** (Opay, M-Pesa, PayPal, Flutterwave).
- ✓ **Investment and trading platforms** (Robinhood, Bamboo, Binance).

4. **E-Commerce and Business Services**

- ✓ **Online marketplaces** (Amazon, Jumia, eBay, Alibaba).
- ✓ **Online booking platforms** (Airbnb, Booking.com, Uber, Bolt).
- ✓ **Business tools** (Shopify for e-commerce, Salesforce for CRM).

CATEGORIES OF ONLINE SERVICES

5. **Educational Services**

- ✓ **Online courses** (Coursera, Udemy, edX, Khan Academy).
- ✓ **Virtual classrooms** (Google Classroom, Moodle, Canvas).
- ✓ **Research tools** (Google Scholar, JSTOR, ResearchGate).

BENEFITS OF ONLINE SERVICES

- ✓ **Convenience** – accessible anytime, anywhere.
- ✓ **Cost-effectiveness** – reduces travel and transaction costs.
- ✓ **Speed** – faster communication, transactions, and learning.
- ✓ **Global reach** – connects users beyond physical boundaries.
- ✓ **Access to Information** – vast knowledge and resources at one's fingertips.
- ✓ **Productivity** – enhances business, learning, and personal efficiency.

CHALLENGES OF ONLINE SERVICES

- ✓ **Security Risks** – phishing, hacking, and online fraud.
- ✓ **Misinformation** – spread of fake news or misleading data.
- ✓ **Addiction and Overuse** – excessive time on social media/gaming.
- ✓ **Privacy Concerns** – personal data misuse.
- ✓ **Digital Divide** – unequal access to online services due to poor connectivity or lack of skills.

2.6: CYBER SECURITY AND INTERNET SAFETY

- ✓ **Cyber Security** refers to the practice of protecting computer systems, networks, digital devices and online data from unauthorized access, attacks, theft, or damage. It involves the use of tools, policies, and safe practices to ensure that information shared or stored online remains confidential, secure and trustworthy.
- ✓ **Internet Safety** (also called Online Safety or Cyber Safety) is the responsible use of the Internet to protect oneself and others from risks such as fraud, cyberbullying, identity theft, and exposure to harmful content. It focuses on individual awareness, behavior, and precautions while online.

KEY CONCEPTS IN CYBER SECURITY

- ✓ **Confidentiality** – Ensuring that information is accessible only to authorized persons.
- ✓ **Integrity** – Protecting information from being altered or corrupted.
- ✓ **Availability** – Ensuring data and services are available whenever needed.
- ✓ **Authentication** – Verifying the identity of users (e.g., passwords, biometrics).
- ✓ **Authorization** – Granting users access only to the resources they are permitted to use.

COMMON CYBER THREATS

- ✓ **Phishing** – Fraudulent emails or messages that trick users into giving out personal information.
- ✓ **Malware** – Malicious software like viruses, worms, and Trojans that damage or steal data.
- ✓ **Ransomware** – A type of malware that locks files until a ransom is paid.
- ✓ **Identity Theft** – Stealing someone's personal information for fraudulent use.
- ✓ **Hacking** – Unauthorized access to computer systems or accounts.
- ✓ **Denial of Service (DoS) Attacks** – Overloading a system to make it unavailable.
- ✓ **Cyberbullying** – Online harassment through social media, messaging, or email.

PRINCIPLES OF INTERNET SAFETY

- ✓ Strong Passwords – Use complex, unique passwords and update them regularly.
- ✓ Two-Factor Authentication (2FA) – Adding an extra layer of security (e.g., OTPs).
- ✓ Safe Browsing Habits – Avoiding suspicious links, downloads, and unsecured websites.
- ✓ Privacy Protection – Limiting what personal information is shared online.
- ✓ Regular Updates – Keeping software, browsers, and antivirus up to date.
- ✓ Secure Networks – Using trusted Wi-Fi connections, preferably encrypted ones.
- ✓ Backup of Data – Regularly saving files to cloud storage or external drives.

Cyber Security Tools and Measures

- ✓ **Antivirus and Anti-Malware Software** (e.g., Avast, Kaspersky, Windows Defender).
- ✓ **Firewalls** – Preventing unauthorized access to networks.
- ✓ **Encryption** – Protecting sensitive data using secure coding.
- ✓ **Virtual Private Networks (VPNs)** – Securing communication by masking IP addresses.
- ✓ **Parental Control Tools** – Protecting children from harmful or inappropriate content.

2.7: PRACTICING SAFE BROWSING

❑ Definition

- ✓ *Safe Browsing* refers to the practice of using the Internet in a secure, responsible and cautious manner to protect oneself from threats such as viruses, scams, identity theft, malware and inappropriate content.
- ✓ It involves *adopting good browsing habits*, *security tools* and *awareness strategies* while exploring websites and online platforms.

Importance of Safe Browsing

1. *Protects against cyber threats* such as phishing, malware and hacking.

2. *Safeguards* personal data and financial information.

3. *Promotes healthy online behavior*, especially for young users.

4. *Prevents exposure* to inappropriate or harmful content.

5. *Builds trust* in online learning, e-commerce, and digital collaboration.

Safe Browsing Practices

1. Use Secure Websites – Look for “https://” and a padlock symbol before entering personal or financial details.

2. Avoid Suspicious Links – Do not click on unknown or shortened links in emails, ads, or messages.

3. Update Browsers Regularly – Ensure Chrome, Firefox, Edge, or Safari are up-to-date with security patches.

4. Install Browser Security Extensions – Use ad blockers, anti-phishing tools, and script blockers.

5. Clear Cookies and Cache Frequently – Prevents tracking and improves security.

Importance of Safe Browsing

6. Beware of Free Downloads – Only download apps, music, or software from trusted sources.

7. Use Strong and Unique Passwords – Avoid reusing the same password across multiple sites.

8. Log Out of Shared Devices – Especially after using cybercafés, libraries, or public computers.

9. Check Privacy Settings – On social media and other platforms to control who sees your information.

10. Think Before Sharing – Avoid oversharing personal details like addresses, phone numbers